JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO AGS SHANTHI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

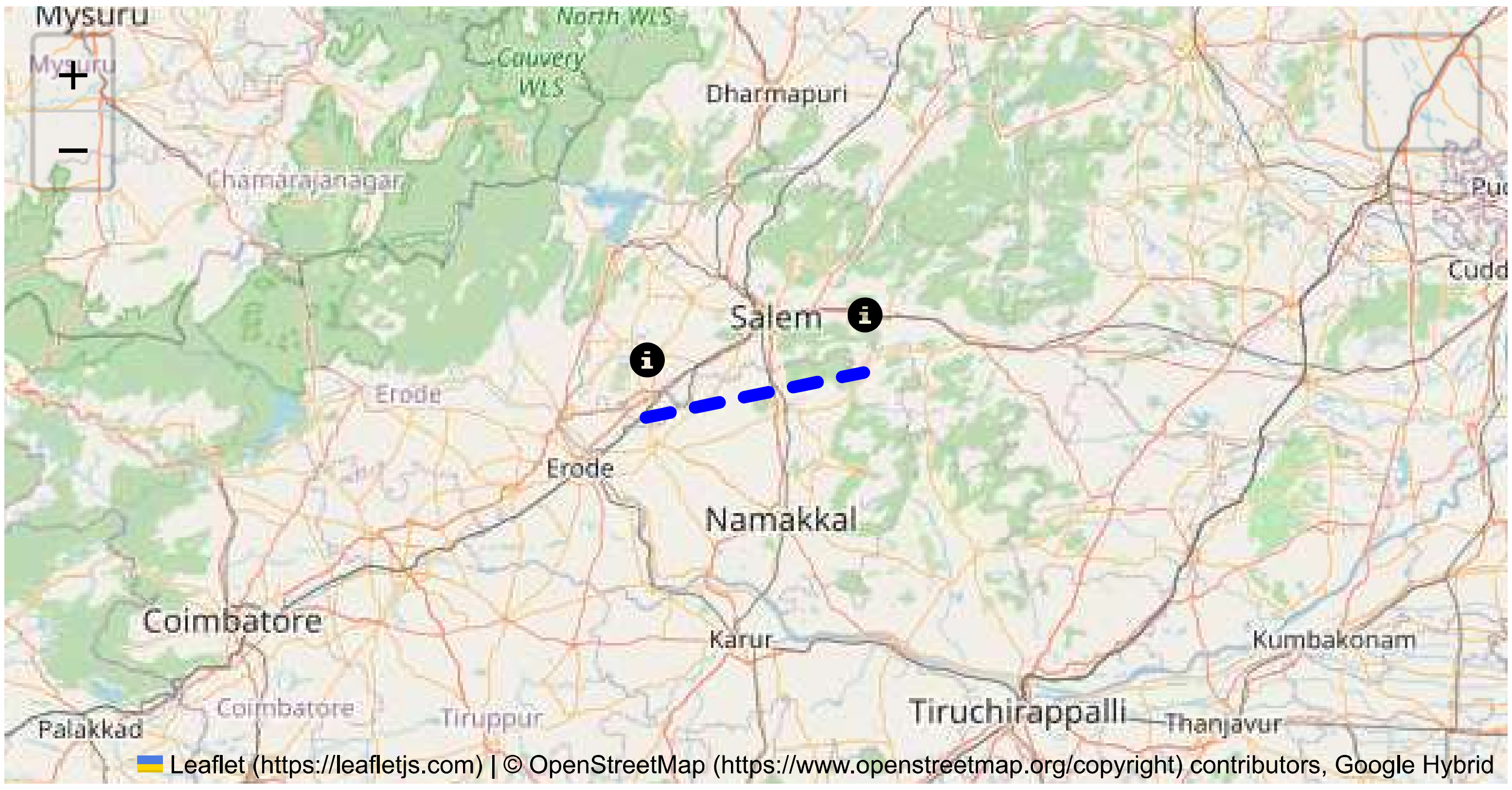
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

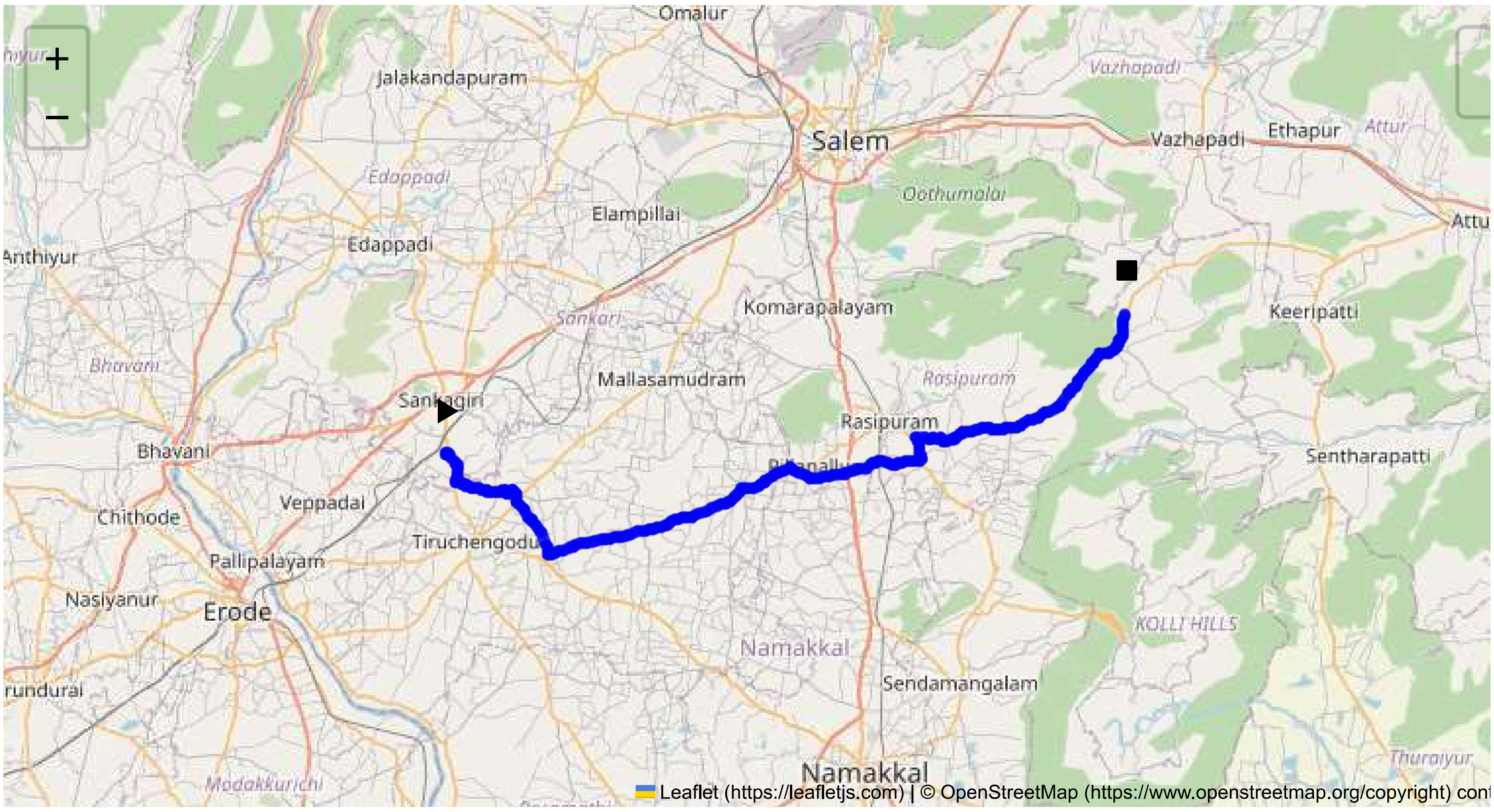
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 67.42 km
Estimated Duration: 1.5 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours
Start: (11.4381, 77.8734)
End: (11.535207, 78.350369)



Welcome to the Journey Risk Management Study

Sure, here's a detailed analysis of the route from CVQF+23W, Sangagiri, Tamil Nadu to G8PX+5P, Pudupatti R.F., Tamil Nadu, focusing on potential hazards:

1. **Overview of the Route Map**:
    - The route spans approximately 67.42 kilometers.

- The journey typically involves traveling on NH544, a major highway, and local roads.
- It passes through both urban and rural areas, crossing several smaller towns and villages.

2. **Typical Weather Conditions and Potential Weather-Related Hazards**:
    - The region experiences a tropical climate with hot summers, moderate winters, and substantial rainfall during the monsoon (June to September).
    - Potential hazards include heavy rain and flooding during monsoon months, which might affect visibility and road conditions, especially in low-lying areas.
3. **Analysis of Traffic Patterns**:
    - Peak congestion is typically observed in the morning (8:00-10:00 AM) and evening (5:00-7:00 PM).
    - Traffic is denser near urban centers and smaller towns.
    - Vehicle flow may be disrupted by agricultural transports and local markets on rural roads.
4. **Assessment of Road Quality and Infrastructure**:
    - NH544 is generally well-maintained, appropriate for heavy vehicles, with multiple lanes and adequate signage.
    - Local roads may vary in quality, with some sections being narrow, poorly maintained, or under construction, affecting travel time and safety.
5. **Alternative Routes for Emergencies**:
    - In case of roadblocks or accidents, alternative routes include using smaller state highways that run parallel to NH544.
    - Detours should be checked for weight restrictions before use, as smaller bridges and roads may not support heavy truck loads.
6. **Summary of Local Regulations Affecting Hazardous Material Transport**:
    - Transport of hazardous materials is subject to national guidelines, including restrictions on carrying certain materials through populated areas.
    - Checkpoints may enforce documentation and safety standards compliance, especially regarding load and vehicle conditions.
7. **Overview of Historical Incidents**:
    - There have been past reports of heavy vehicle accidents, mainly due to mechanical failures and poor visibility.
    - Historical data might indicate areas prone to such incidents, often near busy intersections and sharp bends.
8. **Environmental Considerations and Sensitive Areas**:
    - Pay attention to wildlife zones and protected areas, particularly around forested or rural segments.
    - Pollution controls are stricter near water bodies to prevent contamination.
9. **Analysis of Communication Coverage**:
    - Cellular and GPS signal strength is generally good along NH544 but can be patchy in rural and mountainous areas.

- Consider offline maps and radios for communication in potential dead zones.

10. **Estimated Emergency Response Times**:

    - In urban regions and close to major towns, emergency response times are generally quicker (within 30 minutes).
    - Rural and remote segments may take longer (up to 1-2 hours) for emergency services to reach affected areas.

11. **Overall Summary of Risk Assessment**:

    - While the main highway facilitates relatively safer travel for heavy vehicles, several challenges persist due to fluctuating weather, road quality, and traffic patterns.
    - Drivers should prepare for possible delays and detours, maintain communication tools, and adhere strictly to safety regulations regarding hazardous materials.
    - Constant monitoring of local news and real-time traffic updates can help mitigate risks.
    - Comprehensive safety training for drivers focusing on emergency protocols, especially in adverse conditions, is highly recommended.

This analysis aims to enhance safety awareness and preparedness for truck drivers carrying hazardous materials on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	High	11.41981, 77.88078	15 KM/Hr
4	Turn	Medium	11.41885, 77.88455	30 KM/Hr
5	Turn	Medium	11.41829, 77.88568	30 KM/Hr
6	Turn	High	11.41859, 77.88692	15 KM/Hr
7	Turn	Medium	11.41761, 77.88739	30 KM/Hr
8	Turn	Medium	11.41716, 77.88935	30 KM/Hr
9	Turn	Medium	11.41603, 77.89479	30 KM/Hr
10	Turn	Medium	11.41439, 77.90025	30 KM/Hr
11	Turn	High	11.41377, 77.90073	15 KM/Hr
12	Turn	Medium	11.41375, 77.90128	30 KM/Hr
13	Turn	Medium	11.41389, 77.90161	30 KM/Hr
14	Turn	Medium	11.41325, 77.90313	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Turn	Medium	11.41276, 77.90349	30 KM/Hr
16	Turn	High	11.41230, 77.90485	15 KM/Hr
17	Turn	Medium	11.41262, 77.90506	30 KM/Hr
18	Turn	Medium	11.41273, 77.90524	30 KM/Hr
19	Turn	Medium	11.41329, 77.90978	30 KM/Hr
20	Turn	High	11.41424, 77.91433	15 KM/Hr
21	Turn	Medium	11.41409, 77.91458	30 KM/Hr
22	Turn	High	11.41208, 77.91551	15 KM/Hr
23	Turn	Medium	11.41218, 77.91651	30 KM/Hr
24	Turn	Medium	11.41186, 77.91768	30 KM/Hr
25	Turn	Medium	11.41342, 77.92087	30 KM/Hr
26	Turn	Medium	11.41341, 77.92113	30 KM/Hr
27	Turn	Medium	11.41319, 77.92144	30 KM/Hr
28	Turn	Medium	11.40833, 77.92362	30 KM/Hr
29	Turn	Medium	11.40669, 77.92312	30 KM/Hr
30	Turn	Medium	11.40566, 77.92355	30 KM/Hr
31	Turn	High	11.40542, 77.92346	15 KM/Hr
32	Blind Spot	Blind Spot	11.36920, 77.94568	10 KM/Hr
33	Turn	Medium	11.38968, 78.02925	30 KM/Hr
34	Turn	Medium	11.41343, 78.08073	30 KM/Hr
35	Turn	High	11.41466, 78.08070	15 KM/Hr
36	Turn	High	11.42962, 78.11578	15 KM/Hr
37	Turn	Medium	11.42450, 78.15278	30 KM/Hr
38	Turn	Medium	11.42953, 78.16850	30 KM/Hr
39	Turn	High	11.43542, 78.20754	15 KM/Hr
40	Blind Spot	Blind Spot	11.45100, 78.20385	10 KM/Hr
41	Turn	Medium	11.44994, 78.22138	30 KM/Hr
42	Turn	Medium	11.44919, 78.22291	30 KM/Hr
43	Turn	High	11.44806, 78.22308	15 KM/Hr
44	Turn	Medium	11.49813, 78.32398	30 KM/Hr
45	Turn	Medium	11.50501, 78.33083	30 KM/Hr
46	Turn	Medium	11.50776, 78.33301	30 KM/Hr
47	Turn	Medium	11.50861, 78.33957	30 KM/Hr
48	Turn	Medium	11.51854, 78.34807	30 KM/Hr